

CS104: Computer Networks & Internet Protocol Architectures

Department of Computer Science & Engineering | Academic Handout

Page 1: Layered Architectures & Transport Protocols (TCP vs UDP)

4.1 Layered Network Models: OSI 7-Layer vs TCP/IP 4-Layer

Network architectures rely on layered abstractions to ensure modular protocol design:

- OSI 7-Layer Model: 7. Application, 6. Presentation, 5. Session, 4. Transport, 3. Network, 2. Data Link, 1. Physical.
- TCP/IP 4-Layer Model: 4. Application (HTTP, DNS, SMTP), 3. Transport (TCP, UDP), 2. Internet (IP, ICMP, ARP), 1. Link / Network Interface (Ethernet, Wi-Fi).

Encapsulation wraps headers at each descending layer: Application Data -> Transport Segment -> Network Packet/Datagram -> Link Frame -> Physical Bits.

4.2 Transport Layer Protocols: TCP versus UDP

Transmission Control Protocol (TCP): Connection-oriented, full-duplex protocol providing reliable, ordered, and error-checked byte stream delivery between applications. Implements sliding-window flow control and dynamic congestion control. Header size ranges from 20 to 60 bytes.

User Datagram Protocol (UDP): Lightweight, connectionless transport protocol providing best-effort, unordered, and unreliable datagram transmission without flow or congestion control. UDP header is fixed at 8 bytes (Source Port, Destination Port, Length, Checksum). Ideal for latency-sensitive applications such as VoIP, online gaming, and DNS queries where packet loss is preferable to retransmission latency.

Page 2: TCP Connection Lifecycle, Flow Control & Congestion Management

4.3 TCP 3-Way Handshake and 4-Way Termination

To establish a reliable connection, TCP executes a Three-Way Handshake:

1. SYN: Client chooses an Initial Sequence Number (ISN) x and sends a segment with SYN flag set: SYN=1, Seq= x .
2. SYN-ACK: Server receives SYN, allocates buffers, selects ISN y , and responds: SYN=1, ACK=1, Seq= y , Ack= $x+1$.
3. ACK: Client acknowledges the server's sequence number: ACK=1, Seq= $x+1$, Ack= $y+1$. Connection transitions to ESTABLISHED.

Connection teardown uses a Four-Way Handshake:

1. Client sends FIN.
2. Server sends ACK.
3. Server finishes sending remaining data and transmits FIN.
4. Client replies with ACK and enters TIME_WAIT state (waiting 2 MSL - Maximum Segment Lifetime) before closing to ensure the final ACK was received.

4.4 Flow Control and Congestion Control

Flow Control prevents a fast sender from overwhelming a slow receiver's buffer space. The receiver continuously advertises its available buffer capacity in the TCP header as the Receiver Window (rwnd). The sender guarantees that $(\text{LastByteSent} - \text{LastByteAcked}) \leq \text{rwnd}$.

Congestion Control prevents senders from overwhelming intermediate network links and routers:

- Slow Start: Sender begins with Congestion Window $\text{cwnd} = 1$ MSS and doubles cwnd exponentially every round-trip time (RTT) until reaching the slow start threshold (ssthresh).
- Congestion Avoidance: Once $\text{cwnd} \geq \text{ssthresh}$, cwnd increases linearly by 1 MSS per RTT (Additive Increase).
- Fast Retransmit and Fast Recovery: Upon receiving 3 duplicate ACKs, TCP retransmits the missing segment immediately without waiting for retransmission timer expiration, halves ssthresh , and resumes linear growth (Multiplicative Decrease - AIMD).

Page 3: DNS Architecture, HTTP Evolution & TLS 1.3 Handshake

4.5 Domain Name System (DNS) Resolution Hierarchy

DNS provides a globally distributed, hierarchical database translating human-readable hostnames into IP addresses:

1. Root DNS Servers: Top of the hierarchy (13 root server IP clusters designated a.root-servers.net to m.root-servers.net).
2. Top-Level Domain (TLD) Servers: Manage specific top-level domains (.com, .org, .edu, .gov, country codes).
3. Authoritative Nameservers: Contain the definitive mapping records for an organization's domain.

Common DNS Record Types: A (maps hostname to IPv4 address), AAAA (maps hostname to IPv6 address), CNAME (canonical name alias), MX (mail exchange server), TXT (arbitrary text used for SPF/DKIM verification).

Recursive queries require the local DNS resolver to perform all external queries on behalf of the client, whereas iterative queries return referral addresses to the next server in the hierarchy.

4.6 HTTP Protocols and TLS 1.3 Security

- HTTP/1.1: Introduced persistent connections (Keep-Alive), but suffered from Head-of-Line (HoL) blocking at the application layer.
- HTTP/2: Replaced text protocols with binary framing. Multiplexes multiple concurrent streams across a single TCP connection, eliminating application-level HoL blocking. Utilizes HPACK header compression.
- HTTP/3: Operates over QUIC (Quick UDP Internet Connections) using UDP as the transport layer. Solves TCP-level HoL blocking caused by packet loss and provides zero round-trip (0-RTT) connection resumption.
- TLS 1.3 Cryptographic Handshake: Modern security standard reducing handshake latency to 1 Round Trip Time (1-RTT). Eliminates insecure cipher suites and mandates ephemeral Diffie-Hellman key exchange (ECDHE) for forward secrecy.